Data Science Examination

Part I: Multiple Choice Questions (1 Mark Each)

Instructions: Select the most appropriate answer from the given options.

1. Which of the following is an assumption of Linear Regression?

A. High multicollinearity   B. Little or no autocorrelation   C. Heteroscedasticity   D. All of the mentioned

2. In a perfectly fitted regression model, the sum of residuals (errors) should be:

A. Maximum   B. Minimum   C. Zero   D. Unknown

3. Which distance metric is specifically used for categorical variables or strings of equal length?

A. Hamming Distance   B. Euclidean Distance   C. Manhattan Distance   D. Minkowski Distance

4. Supervised learning can be split into which two main categories?

A. Classification   B. Regression   C. Both A & B   D. None of the mentioned

5. Which of the following is an ensemble learning method based on decision trees?

A. Support Vector Machine   B. Random Forest   C. K-Nearest Neighbors   D. Logistic Regression

6. The process of combining multiple classifiers to improve prediction (e.g., Bagging or Boosting) is called:

A. Voting   B. Clustering   C. Dimensionality Reduction   D. Feature Engineering

7. In a Confusion Matrix, what is the value of the error for a perfect classifier?

A. Zero   B. One   C. 0.5   D. Depends on the dataset

8. Which visualization tool is most effective for identifying correlations between two continuous variables?

A. Bar Chart   B. Box Plot   C. Scatter Plot   D. Histogram

9. In a scatter plot, individual data observations are represented as:

A. Points   B. Lines   C. Bars   D. Areas

10. Which measure of central tendency is most sensitive to outliers?

A. Median   B. Mean   C. Mode   D. Range

11. In regression analysis, the output (dependent variable) is typically:

A. A category   B. A real number   C. A boolean   D. An image

12. The process of identifying the most relevant variables to use in a model is known as:

A. Data cleaning   B. Column dropping   C. Feature selection   D. Normalization

13. Which technique is used to predict a continuous numerical value?

A. Classification   B. Regression   C. Clustering   D. Association

14. Analysis that involves data points collected or recorded at specific time intervals is called:

A. Linear Analysis   B. Binary Analysis   C. Time Series   D. Discrete Analysis

15. Which of the following is actually a classification algorithm despite its name?

A. Linear Regression   B. Logistic Regression   C. Multiple Regression   D. Polynomial Regression

16. Hyperparameters are best described as:

A. Data labels   B. Input features   C. Parameters of the ML algorithm set before training   D. The final weights of a model

17. (True/False): Logistic regression is used for predicting continuous outcomes.

A. True   B. False

18. In a confusion matrix, where are the "Correct Predictions" located?

A. In the first row   B. In the upper diagonal   C. Along the main diagonal   D. In the last column

19. What is the formula for Precision?

A. TP / (TP + FN)   B. TP / (TP + FP)   C. TN / (TN + FP)   D. (TP + TN) / Total

20. In Support Vector Machines (SVM), the data points closest to the hyperplane that help define the margin are called:

A. Outliers   B. Support vectors   C. Clusters   D. Centroids

Part II: Short Answer Questions (2 Marks Each)

21. Define Supervised Learning.

22. Evaluation Case Study: A model (M1) achieves 99% accuracy on the training set but only 30% accuracy on the testing set. What is happening to this model?

23. What is Regression Analysis?

24. List and elucidate two major challenges in Machine Learning.

25. List any two major real-world applications of Data Science.

26. Distinguish between Structured and Semi-structured data.

27. Describe methods used to detect and handle outliers in a dataset.

28. What is Data Pre-processing?

29. Why is Data Normalization required for the K-Nearest Neighbors (K-NN) algorithm?

30. How are continuous attributes discretized? Give an example.

31. What does the 'K' represent in the K-Means clustering algorithm?

32. Name three common Hyperparameters in Machine Learning algorithms.

33. Define Gradient Descent.

34. Classifier Comparison: Compare three SVM classifiers based on their margins and boundaries.

35. Explain how the Radial Basis Function (RBF) Kernel works in SVM.